

Simplify the following expressions:

$$a) e^{\ln 5}$$

$$\Rightarrow e^{\ln 5} = 5$$

$$b) e^{2\ln 3} + e^{-\ln 2}$$

$$e^{2\ln 3} + e^{-\ln 2} = 9 + \frac{1}{2} = \boxed{\frac{19}{2}}$$

$$c) \frac{e^{2x-3}}{e^{4x-3}}$$

$$\Rightarrow e^{2x-3-(4x-3)} = e^{2x-3-4x+3} = \boxed{e^{-2x}}$$

$$d) (e^x)^3 \cdot e^{-2x}$$

$$\Rightarrow e^{3x} \cdot e^{-2x} = e^{3x+(-2x)} = \boxed{e^x}$$

$$e) \frac{e^{\ln 3}}{\ln e^9}$$

$$\Rightarrow \frac{3}{9} = \boxed{\frac{1}{3}}$$

$$f) \ln(e^{2+\sqrt{3}} \times e^{2-\sqrt{3}})$$

$$\Rightarrow \ln(e^{2+\sqrt{3}}) + \ln(e^{2-\sqrt{3}})$$

then:

$$2+\sqrt{3} + 2-\sqrt{3} = \boxed{2}$$

$$g) \sqrt{\frac{2 \ln \sqrt[3]{e}}{e^{\ln 3 - \ln 2}}}$$

$$\Rightarrow \sqrt{\frac{\ln \sqrt[3]{e}^2}{e^{\ln(\frac{3}{2})}}} = \sqrt{\frac{\frac{2}{3} \ln \sqrt[3]{e}}{e^{\ln(\frac{3}{2})}}}$$

$$= \sqrt{\frac{\frac{2}{3} \ln \sqrt[3]{e}}{\ln(e^{\frac{3}{2}})}} = \sqrt{\frac{\ln \sqrt[3]{e}^2}{\ln(e^{\frac{3}{2}})}}$$

$$\Rightarrow \sqrt{\frac{\frac{2}{3}}{\frac{3}{2}}} = \sqrt{\frac{2}{3} \left(\frac{2}{3} \right)} = \sqrt{\frac{4}{9}}$$

$$\Rightarrow \frac{\cancel{2}}{\cancel{3}} = \boxed{\frac{2}{3}}$$

Day 7

[Signature]